

CS549.V Midterm #1

I. QUESTION (8 POINTS)

Write regular expressions for the following items:

- A price in Turkish lira less than 1000 that also includes the kuruş part. Examples: 1.3 TL, 12.64TL, 123.90TL
- All words that contains at least one Turkish character: ç, ö, ğ, ü, ş, ı. Examples: örgü, ığdır, çöğüş, ağ

II. QUESTION (12 POINTS)

Draw a simple finite state machine (FSM) for English language where

- FSM has start states for nouns and verbs.
- Supports plural forms for nouns.
- Supports simple present, past and simple continuous tenses.

III. QUESTION (10 POINTS)

Give example Turkish words for the following:

- The word has only one morphological analysis.
- The word has more than one morphological analysis where there is only one possible root word.
- The word has more than one morphological analysis where there are more than one possible root word.
- The morphological analyses of the word have at least one noun and one verb root.
- One morphological analysis of the word starts with verb root and then has a derivation boundary to noun and also is in dative form.

IV. QUESTION (18 POINTS)

- Construct a mini-corpus such that, the following bigram probabilities (without Laplace smoothing) holds:

$$P(I \mid \langle s \rangle) = 2 / 4$$

$$P(\text{Sam} \mid \langle s \rangle) = 1 / 4$$

$$P(\text{am} \mid I) = 2 / 5$$

$$P(\langle /s \rangle \mid \text{Sam}) = 1 / 3$$

$$P(\text{Sam} \mid \text{am}) = 1 / 2$$

$$P(\text{do} \mid I) = 1 / 5$$

- Calculate the same bigram probabilities with Laplace smoothing.

V. QUESTION (20 POINTS)

Let us say we have an hidden Markov model with 2 states HOT and COLD; and three possible words in the output vocabulary 1, 2, 3.

with initial probabilities: $\Pi = [0.8, 0.2]$

the transition matrix:

$$\mathbf{A} = \begin{vmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{vmatrix}$$

the emission probabilities:

$$B_{HOT} = \begin{vmatrix} 0.2 \\ 0.4 \\ 0.4 \end{vmatrix}$$

$$B_{COLD} = \begin{vmatrix} 0.5 \\ 0.4 \\ 0.1 \end{vmatrix}$$

- Draw the HMM.
- Calculate forward probability for the observation sequence 1, 2, 3 using the forward algorithm.
- Calculate the best hidden weather sequence for the observation sequence 1, 2, 3 using the Viterbi algorithm.

VI. QUESTION (8 POINTS)

Given the following contingency table, calculate the precision, recall, F-measure and accuracy scores.

	Gold positive	Gold negative
System positive	70	30
System negative	40	60

VII. QUESTION (12 POINTS)

Write pseudocode to implement K-fold cross-validation.

VIII. QUESTION (12 POINTS)

Derive the update equation for the logistic regression for the cross-entropy loss function, where the sigmoid is changed with the hyperbolic tangent function

$$G(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}} \quad (1)$$

Remember the cross-entropy loss for sigmoid function is defined as:

$$L(w, b) = -[y \log \sigma(wx + b) + (1 - y) \log(1 - \sigma(wx + b))] \quad (2)$$